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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	505 Server Error
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth & SDG 9: Industry, Innovation and Infrastructure
Project / Solution Name	NGate
Name (Leader)	Melissa Ting Jia Jie
Email & Phone (Leader)	mtjiajie21@gmail.com & 010-9665679
University Affiliation(s)	Universiti Malaysia Perlis (UniMAP)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian independent mini-marts face a practical purchasing problem: the owner must decide what to buy, how much to order and which supplier to trust, often using scattered WhatsApp messages, PDF quotations and incomplete stock records. Malaysia had 1,292,580 MSMEs in 2025, but NGate specifically targets inventory-intensive small retailers rather than claiming every MSME as a customer [1].

Three connected risks affect daily operations. First, repeated purchases from familiar suppliers can hide price changes or missed discounts when older quotations are not checked. Second, buying too much ties up cash and may lead to expiry, while buying too little risks empty shelves and lost sales. Third, comparing unfamiliar product descriptions, minimum order quantities (MOQ), delivery terms and total costs across quotations takes staff time and invites errors.

The cheapest unit price is not always the best purchasing choice. A lower-priced supplier may require excessive quantities or deliver after available stock runs out. Owners need a decision that considers price, quantity and timing together—not another system that merely records the purchase afterward.

NGate addresses this decision gap through an explainable, owner-controlled procurement workflow. Krukkruk Mini Mart is a simulated test retailer, using team-created quotations rather than presenting invented losses as observed Malaysian industry results.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

NGate is an AI-assisted purchasing and stock decision-support prototype for small retailers. It combines document extraction, transparent cost calculations, stock indicators and explicit human approval in a guided five-step workflow.

1. MONITOR — A dashboard shows products, stock-on-hand, reorder thresholds, estimated days of stock cover and supplier lead-time indicators. Current figures use seeded demonstration records, not a live cashier/POS feed.

2. REQUEST — Staff select an item and NGate prepares an RFQ message. Users open WhatsApp or copy the text and send it themselves; NGate does not contact suppliers automatically.

3. READ & COMPARE — Staff upload supplier PDF/JPG quotations (up to 8 MB). Google Gemini extracts available prices, quantities, MOQ, order multiples and delivery details; missing or uncertain values are flagged for review. Confirmed quotations are then compared by rule-based backend calculations covering total spend, discounts, MOQ-related surplus and delivery suitability. The extraction output currently requires human transfer into the purchasing record; it is not an automatic end-to-end connection.

4. APPROVE — The owner examines the alternatives, selects a supplier and explicitly approves the decision. NGate can produce a downloadable purchase-order (PO) draft; sending it remains a manual action.

5. RECORD — The system retains purchasing and activity records. Actual POS synchronisation and receipt-derived supplier lead-time learning are planned, not yet live.

The demonstration uses nine team-created supplier quotation PDFs covering 22 products. The React/Vite interface, Express API, Flask purchasing engine, databases and Gemini integration run through Replit. Its modular design translates the SDC workshops' GenAI, data-architecture and cloud-deployment skills into a functioning prototype without removing owner control.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

NGate uses a modular architecture with distinct responsibilities:

• Presentation: React and Vite provide the SME dashboard, product search, quotation views and guided purchasing interface. The primary presentation currently uses mock data.

• Frontend-facing service: Node.js, TypeScript and Express handle authenticated requests, Replit OpenID Connect sessions, quotation-file uploads and a protected read-only bridge to Flask.

• AI document service: Google Gemini extracts structured quotation fields from user-uploaded PDF/JPG files. The extracted values are validated and shown for user review; Gemini does not approve purchases.

• Purchasing service: Python and Flask provide stock indicators, manual RFQ drafts, recorded quotation comparison, owner selection, PO status and PDF generation through ReportLab. Comparisons use deterministic business logic.

• Storage: PostgreSQL stores user/session/upload metadata; Replit private object storage contains quotation files; SQLite stores the Krukkruk Mini Mart demo business and purchasing records. Drizzle ORM accesses PostgreSQL.

• Hosting: Replit supports development, integrated service execution and prototype hosting through the SDC sponsor resource.

Implemented data path: React/Vite → Express → Gemini/storage and authenticated read-only Flask bridge → SQLite. The Flask purchasing workflow is not yet the React app's primary live data source. A future POS connection would supply real sales and stock transactions; it is not claimed as implemented. See Appendix A.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our design process starts with one real purchasing decision: a mini-mart owner must know whether stock can last until the next delivery, whether a supplier quotation is comparable, and whether its total cost is worthwhile. We translated these needs into a compact dashboard, an RFQ-to-PO journey and explicit owner approval rather than attempting autonomous ordering.

Feasibility is demonstrated using nine team-authored PDF quotations and a 22-product seeded purchasing scenario. Flask calculates stock-cover warnings, quotation totals and MOQ exposure; Gemini assists with extracting quotation fields. Separate services allow extraction to fail without inventing a supplier recommendation, while uncertain fields require user review. Our React presentation still uses mock data, so the next integration milestone is connecting the complete purchasing workflow to the interface.

For Phase 2, we will run matched manual-versus-NGate tasks, check extraction against known quotation answers, log decision time, and test purchase choices under fixed prices, MOQ and delivery constraints. We will then pilot POS imports and record actual PO-to-receipt events before claiming live demand or lead-time predictions. The owner remains accountable for every purchase.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**PRIMARY — SDG 8, Target 8.2:** Improve economic productivity through technological upgrading and innovation. NGate targets the time and errors involved in small-retailer procurement. **SUPPORTING — SDG 9, Target 9.5:** Strengthen technological capabilities and encourage innovation through an accessible AI-assisted business workflow[3]. **SUPPORTING — SDG 12, Target 12.3:** Reduce retail food waste. Stock-aware purchasing may reduce unnecessary perishable inventory; actual kilograms or units wasted must be measured in a real pilot [4].
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	DEMONSTRATION BASELINE (not an industry loss estimate): A manual user must read nine team-created supplier quotations covering 22 products and three purchasing categories. The manual process has no automatic quotation-field extraction or cost comparison. Measured quotation scenario: purchasing the three sample baskets from the assumed baseline selections costs RM2,531.80; a lower quoted-cost combination totals RM2,418.15. The difference is RM113.65 per complete basket cycle, or 4.49% of the baseline [5]. This is a potential quotation-price gap—not verified SME savings, and it excludes differences in delivery charges and availability. Actual baseline processing minutes, stockouts and spoiled units have not been measured. These must be recorded

Metric Category	Your Response
	during matched manual-versus-prototype tests; demonstration stock indicators use seeded data.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. Purchase cost opportunity (RM and %): (baseline comparable basket cost − selected alternative cost) ÷ baseline cost; verify product equivalence, MOQ, delivery and charges. 2. Procurement task time (minutes and %): time from quotation review to an approved PO, measured on identical tasks manually and with NGate. 3. Quotation extraction accuracy (%): correctly extracted required fields ÷ audited fields, against human-verified PDF ground truth; report missing/uncertain fields. 4. Inventory decision quality: low-stock flag agreement with defined reorder rules (%), simulated excess units and stockout events; actual waste avoided is a later pilot KPI.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	PHASE 2 TEST TARGETS (aspirational, not achieved results): • ≥95% correct extraction of selected quotation fields on labelled test PDFs, with uncertain fields referred for review. • ≥30% reduction in median quotation-to-PO task time versus manual processing on identical scenarios. • Correctly detect 100% of predefined low-stock rule breaches in the controlled demo; aim for ≥20% fewer avoidable excess-stock units under a documented replenishment simulation without additional stockouts. • Demonstrate the RM113.65 (4.49%) quoted basket-cost opportunity under the defined comparison conditions; report any realisable savings separately after fees, supplier availability and other costs are checked. Impact on actual food waste or profitability will only be claimed after a retailer pilot measures those outcomes.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	**Type (code/dataset/model)**	**License**
React ; Vite	UI framework ; build tool	MIT
TypeScript ; Drizzle ORM	Code	Apache-2.0
Node.js + Express	Server runtime and framework	MIT
Python	Code	PSF
Flask	Purchasing REST service	BSD-3-Clause
SQLite ; PostgreSQL	Database engines	Public Domain / PostgreSQL License
ReportLab	Purchase-order PDF generation	BSD-style
Google Gemini	Pre-trained AI service	Google Gemini API Terms
Replit	Development and hosting platform	Replit Terms of Service

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

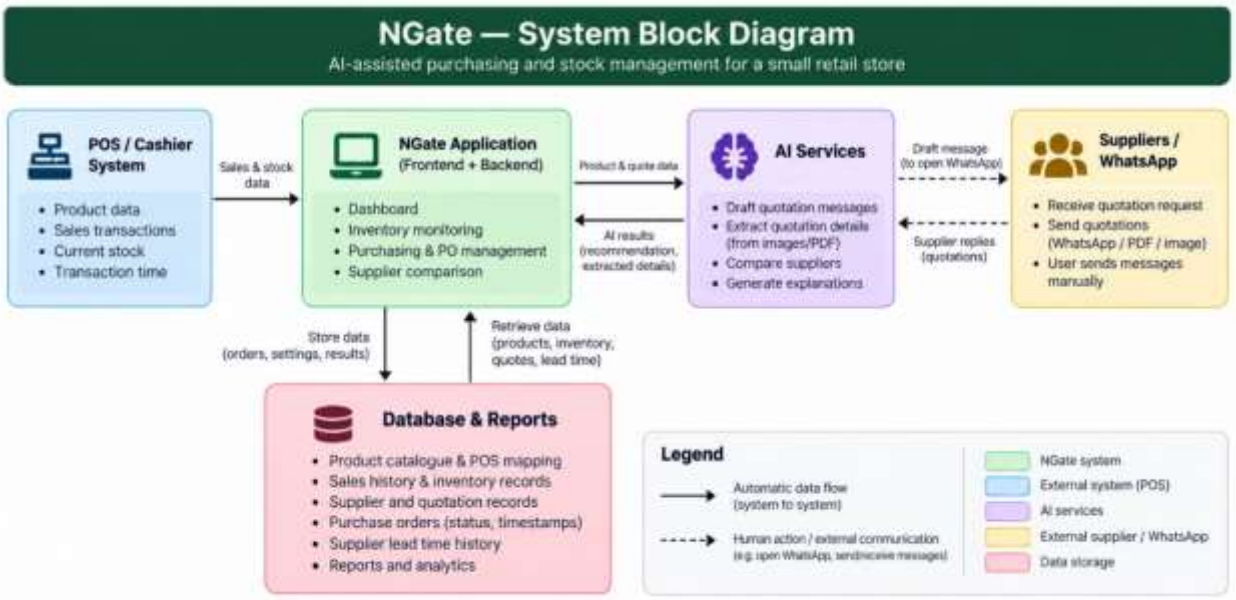
7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24 OCTOBER 2026

Appendix A – NGate System Architecture and POS Integration Plan



Note: POS integration and automated supplier lead-time tracking are planned enhancements.

Appendix B – Supporting Evidence and Financial Calculations

B.1 Data Source and Scope

NGate uses a team-created demonstration dataset consisting of nine supplier quotation PDFs covering 22 products for the simulated Krukkruk Mini Mart.

The quotations support testing of AI-assisted data extraction, supplier comparison, purchasing calculations, and purchase-order workflows.

Data classification: Synthetic demonstration data. The results do not represent verified financial losses or measured savings from an operating Malaysian retailer.

B.2 Supplier Cost Comparison

The following calculation compares the same three purchasing baskets at their specified quoted quantities.

Measurement	**Amount (RM)**
Baseline purchasing cost	2,531.80
Alternative purchasing cost	2,418.15
Potential cost difference	113.65
Potential cost reduction	4.49%

Calculation:

Potential cost difference = RM2,531.80 − RM2,418.15 = RM113.65

Potential cost reduction = (RM113.65 ÷ RM2,531.80) × 100 = 4.49%

Note: These figures represent a quotation-based purchasing opportunity, assuming the compared products and quantities are equivalent. They do not account for additional delivery charges, payment conditions, supplier availability, or other purchasing constraints.

B.3 Illustrative Monthly Financial Impact

Assumption / Calculation	Value
Complete purchasing basket cycles per month	4
Potential saving per cycle	RM113.65
Assumed realisation rate	75%
Estimated monthly purchasing benefit	RM340.95
Proposed NGate Essential subscription	RM99.00/month
Estimated benefit after subscription	RM241.95/month

Calculation:

Monthly purchasing benefit = 4 × RM113.65 × 75% = RM340.95

Benefit after subscription = RM340.95 − RM99.00 = RM241.95/month

Assumptions: Four complete purchasing cycles and a 75% realisation rate are illustrative assumptions, not observed customer behaviour. Actual benefits depend on purchasing frequency, comparable product specifications, supplier availability, delivery charges, and other transaction conditions.

The RM241.95 figure excludes implementation expenses and other customer costs. It is not NGate's operating profit or a guaranteed customer saving.

B.4 References

[1] SME Corporation Malaysia. (2026). *Profile of MSMEs in 2015–2025.* Updated August 2026. https://smecorp.gov.my/index.php/en/policies/2020-02-11-08-01-24/profile-and-importance-to-the-economy

[2] United Nations. (n.d.). *Sustainable Development Goal 8: Decent Work and Economic Growth.* Target 8.2. https://sdgs.un.org/goals/goal8

[3] United Nations. (n.d.). *Sustainable Development Goal 9: Industry, Innovation and Infrastructure.* Target 9.5. https://sdgs.un.org/goals/goal9

[4] United Nations. (n.d.). *Sustainable Development Goal 12: Responsible Consumption and Production.* Target 12.3. https://sdgs.un.org/goals/goal12

[5] NGate Development Team. (2026). *Synthetic Supplier Quotation Dataset and Purchasing Cost Calculations.* Internal demonstration materials, 505 Server Error, Universiti Malaysia Perlis.